Performance Testing

Date	5 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected load and confirms that response times, reliability, and resource usage stay within acceptable limits. Given the project timeline, this was carried out through manual testing, backend test coverage, and code-level design decisions rather than a dedicated load-testing tool run.

Step 1: Testing Overview

Field	Details
Testing Approach Used	Manual endpoint testing, automated backend test suite (pytest), and code-level performance review
Type of Testing	Functional regression testing, input validation testing, manual response-time observation
Target Module / API	/login, /register, /dashboard-data, /financial-health, /settlement-predictor, /ai-negotiation-strategy
Test Environment	Local development server and live deployment (Render backend, Vercel frontend)
Test Date	[FILL IN YOUR TEST DATE]

Step 2: Automated Test Coverage

86 backend tests (pytest) were run to verify correctness and stability across authentication, loan management, financial calculations, settlement prediction, and AI negotiation endpoints, including 15 dedicated input-validation tests. All 86 tests pass.

Step 3: Performance Design Decisions & Manual Observations

S.No	Area	Design Decision / Observation	Status
1	Dashboard response time	Manually observed to load in under 2 seconds locally and on the live Render deployment during testing.	Verified

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2	External AI call handling	Google Gemini API calls are capped at a 15-second timeout, after which the rule-based fallback engine activates automatically — preventing indefinite hangs.	Implemented & Verified
3	Database query performance	Indexes added on all foreign key columns (user_id, loan_id) across the 5 core tables to keep queries fast as data grows.	Implemented
4	Session handling	JWT expiry set to 120 minutes based on manual testing of typical user session length; automatic redirect to login on 401 responses.	Verified
5	Free-tier hosting constraints	Backend runs on Render's free tier (512 MB RAM, shared CPU); acceptable for demo/evaluation traffic, not formally load-tested for concurrent production traffic.	Documented limitation

Step 4: Observations & Analysis

Key Findings: The application performs reliably under normal manual testing and typical single-user interactive load. Dashboard, loan management, and financial calculation endpoints respond quickly since they run entirely server-side without external calls.

Bottlenecks Identified: The primary latency source is the external Google Gemini API call for AI negotiation strategy and letter generation, which can take several seconds depending on network conditions — mitigated by the 15-second timeout and rule-based fallback.

Note on Scope: A formal load-testing tool (e.g. JMeter, Locust) was not run against the deployed environment within the project timeline. This is documented as a limitation rather than represented as completed load-test results. Future work includes running an actual concurrent-user load test before scaling beyond demo/evaluation use.